Linear Programming

Problem 1

A baker makes two types of treats: **Cupcakes (x)** and **Cookies (y)**.

- **Time Constraint:** It takes 2 hours to bake a batch of cupcakes and 1 hour for cookies. He has **8 hours** available.
- **Space Constraint:** He can only fit **5 total batches** in his display case.
- **Objective:** He makes a profit of **$30** per batch of cupcakes and **$20** per batch of cookies. Find the number of cupcake batches (x) and cookie batches (y) that maximize the profit.

optimization problem

x : # of batches of cupcakes

y : # of batches of cookies

Time needed to bake x batches of cupcakes: $2x$

Time needed to bake y batches of cookies: y

In total, time needed would be: $2x + y$

- time constraint: $2x + y \leq 8$
- space constraint: $x + y \leq 5$
- $x \geq 0$, $y \geq 0$

Profit $= 30x + 20y$

we want to maximize the profit.

The Math: Problem 1 (Maximization)

Variables:

- x = Number of batches of Cupcakes
- y = Number of batches of Cookies

Objective Function:
Maximize the total profit (P):

$$P = 30x + 20y$$

Subject to Constraints:

1. **Time:** $2x + y \leq 8$ (Each cupcake batch takes 2 hours; cookies take 1)
2. **Space:** $x + y \leq 5$ (Total batches cannot exceed 5)
3. **Non-negativity:** $x \geq 0, y \geq 0$

Step - by - Step Solution

$\boxed{\text{Step 1}}$: List all the lines | Linear equations.

(A) : $2x + y = 8$

(B) : $x + y = 5$

(C) : $x = 0$

(D) : $y = 0$

$\boxed{\text{Step 2}}$: Find ALL possible intersections between the lines.

(A) and (B):

$$\begin{array}{rl} 2x + y = 8 & (A) \\ - \quad x + y = 5 & (B) \\ \hline x = 3 & \end{array}$$

$x + y = 5 \Rightarrow y = 5 - x \Rightarrow y = 2$

$\Rightarrow \boxed{(x, y) = (3, 2)}$

(A) and (C)

$$\begin{array}{rl} 2x + y = 8 & (A) \\ x = 0 & (C) \end{array}$$

$\Rightarrow y = 8$

$\boxed{(x, y) = (0, 8)}$

(A) and (D)

$$\begin{array}{rl} 2x + y = 8 & \\ y = 0 & \end{array}$$

$\Rightarrow 2x + 0 = 8 \Rightarrow x = 4$

$\Rightarrow (x, y) = (4, 0)$

(x) (B) and (C)

$x + y = 5$ (B)

$x = 0$ (C)

$\Rightarrow \quad y = 5 \quad \Rightarrow \quad \boxed{(x,y) = (0,5)}$

(x) (B) and (D)

$x + y = 5$

$y = 0$ $\quad \Rightarrow \quad (x,y) = (5,0)$

(x) (C) and (D)

$x = 0$

$y = 0$ $\quad \Rightarrow \quad (x,y) = (0,0)$

Step 3 : Check the constraints to find the vertices

x	y	$2x + y \leq 8$	$x + y \leq 5$	Vertices ?
3	2	✓	✓	Yes
0	8	✓	✗	No
4	0	✓	✓	Yes
0	5	✓	✓	Yes
5	0	✗	✓	No
0	0	✓	✓	Yes

Step 4 check to find the max profit

x	y	Profit = 30x + 20y
3	2	130
4	0	120
0	5	100
0	0	0

← max profit

Answer: The baker should make 3 batches of cupcakes and 2 batches of cookies. to maximize the profit.

You try

A small shop builds two types of custom computers: Gaming PCs (x) and Office PCs (y).

- **Time Constraint**: A Gaming PC takes 3 hours to build, and an Office PC takes 1 hour. The shop has 9 hours of building time available today.
- **Component Constraint**: Due to a parts shortage, they can only build a total of 5 computers maximum.
- **Objective**: They earn $50 for every Gaming PC and $30 for every Office PC. Find the number of Gaming PCs (x) and Office PCs (y) that maximize the profit.

Problem 2

An athlete needs to meet daily nutritional requirements using **Protein Shakes (x)** and **Energy Bars (y)**.

- **Protein Constraint:** Each shake has 2 units of protein, and each bar has 1 unit of Protein. He needs at least **5 units of Proteins** a day.
- **Energy Constraint:** Each shake has 1 unit of energy, and each bar has 3 units of Energy. He needs at least **10 units of Energy** a day.
- **Objective:** Each shake costs **$4** and each bar costs **$3**. Find the number of Protein Shakes (x) and Energy Bars (y) that minimize the cost.

Constraints:

1. Protein : $2x + y \geq 5$
2. Energy : $x + 3y \geq 10$
3. $x \geq 0$, $y \geq 0$

Objective : Minimize the cost : $4x + 3y$

The Math: Problem 2 (Minimization)

Variables:

- x = Number of Protein Shakes
- y = Number of Energy Bars

Objective Function:
Minimize the total cost (C):

$$C = 4x + 3y$$

Subject to Constraints:

1. **Protein:** $2x + y \geq 5$
2. **Energy:** $x + 3y \geq 10$
3. **Non-negativity:** $x \geq 0, y \geq 0$

Step 1 : List all the linear Equations

(A) : $2x + y = 5$

(B) : $x + 3y = 10$

(C) : $x = 0$

(D) : $y = 0$

Step 2 : Find All possible intersections

(A) and (B) :

$$2x + y = 5$$
$$x + 3y = 10$$

times 3 to both sides

$$\Leftrightarrow \quad 6x + 3y = 15$$
$$- \quad x + 3y = 10$$
$$5x = 5$$

$$\Rightarrow \quad x = 1$$

plug $x = 1$ into $2x + y = 5$

$$\Rightarrow \quad 2 + y = 5$$

$\Rightarrow \quad y = 3$

$\Rightarrow \quad (x, y) = (1, 3)$

(*) (A) and (C)

$2x + y = 5$

$x = 0$

$\Rightarrow \quad y = 5$

$\Rightarrow \quad (x, y) = (0, 5)$

(*) (A) and (D)

$2x + y = 5$

$y = 0$

$\Rightarrow \quad (x, y) = (2.5, 0)$

(*) (B) and (C)

$x + 3y = 10$

$x = 0$

$\Rightarrow \quad (0, 10/3)$

(*) (B) and (D)

$$x + 3y = 10$$
$$y = 0$$

$\rightarrow (x, y) = (10, 0)$

(*) (C) and (D) $(x, y) = (0, 0)$

<u>Step 3</u>: Check to find the vertices

x	y	$2x + y \geq 5$	$x + 3y \geq 10$	Vertices?
1	3	✓	✓	Yes
0	5	✓	✓	Yes
2.5	0	✓	✗	NO
0	10/3	✗		NO
10	0	✓	✓	Yes
0	0	✗	✗	NO

Step 4

x	y	cost $= 4x + 3y$
1	3	13
0	5	15
10	0	40

minimum cost

Answer: $x = 1$, $y = 3$.

You try

A landscaping company needs to spread mulch and stone using two sizes of trucks: **Small Trucks (x)** and **Large Trucks (y)**.

- **Mulch Capacity Constraint:** A Small Truck carries 1 ton, and a Large Truck carries 3 tons. They need to deliver at least **6 tons** of mulch.
- **Stone Capacity Constraint:** A Small Truck carries 2 tons, and a Large Truck carries 1 ton. They need to deliver at least **7 tons** of stone.
- **Objective:** Running a Small Truck costs **$100** and a Large Truck costs **$200**. Find the number of Small Trucks (x) and Large Trucks (y) that minimize the cost.